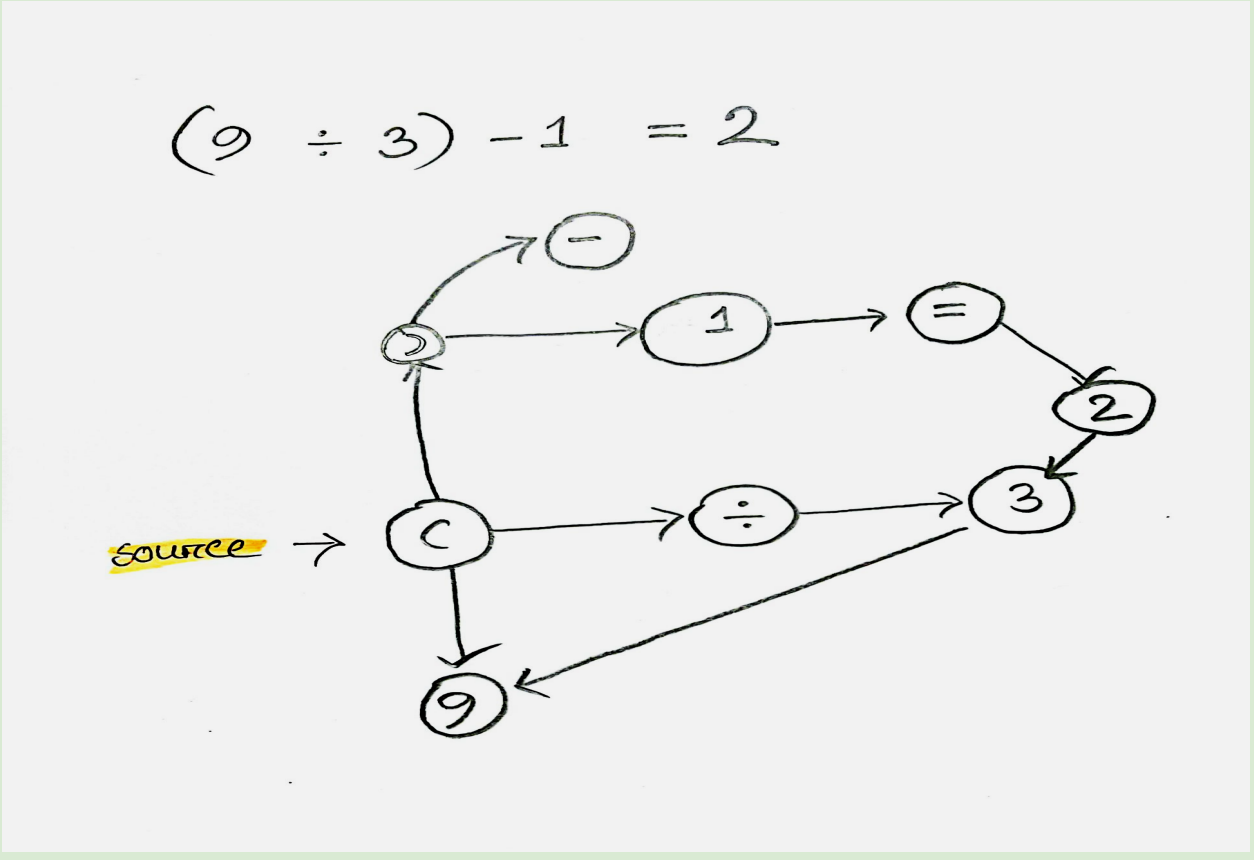
CSE 221 Quiz 02 A

Time:[7:00pm to 7.40pm] 40 minutes (including submission)

Bill says he has found a valid mathematical equation while running DFS from the node denoted by '('. According to him, the equation is:

(9 ÷ 3) - 1 = 2



a.

Compute the number of edges this DFS tree (of the equation) contains. [1]

b.

Is what Bill said right i.e is the equation achievable by running such a DFS?

Validate it by showing steps. (Neighbor/Edge selection should be done according to the necessity of the verification process, You must show start and stop time for each Node while simulating) [5]

CSE 221 Quiz 02 A

Time:[7:00pm to 7.40pm] 40 minutes (including submission)

c.

Classify the edges of the graph into tree edge, back edge and forward edge while running DFS from the node denoted by '('.
[fill the blanks with Yes or No]

[3]

Edge (u→v)	tree edge	back edge	forward edge
"=" –> "2"			
"2" –> "3"			
"÷ " –> "3"			

d.

Explain in 1 or 2 lines, how can you traverse/explore the whole graph by implementing DFS if the graph is disconnected. [1]

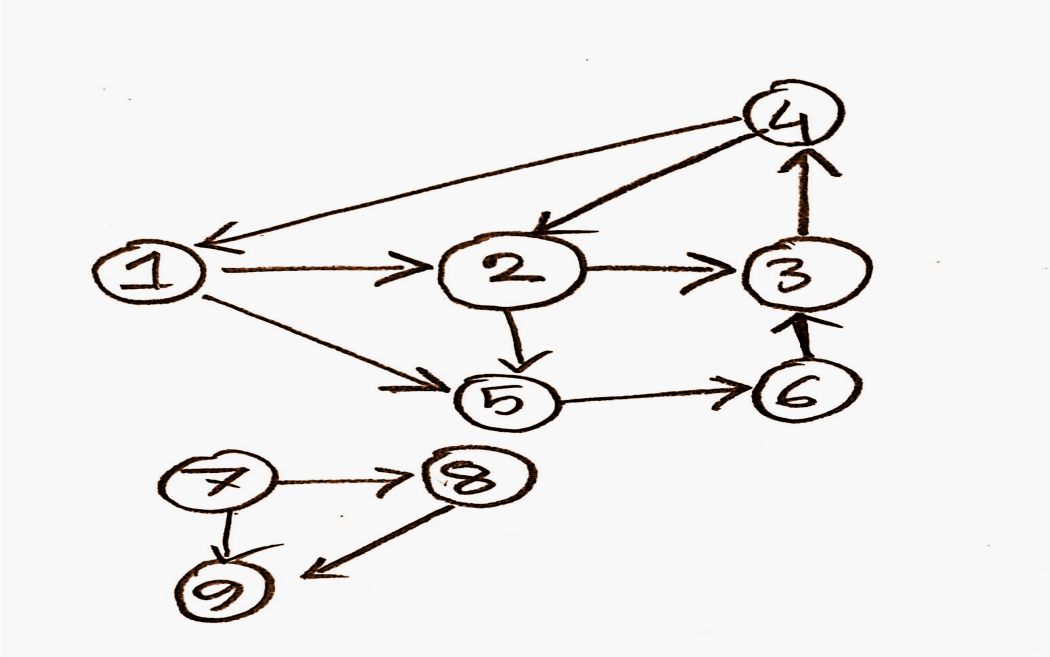
[BONUS] e. If a graph has considerably more adjacent nodes for each node [dense graph], you are given a source point and a destination point. If the destination is at the maximum number of hops away from source, which traversal algorithm will work poorly? [1 line answer] [1]

SUBMISSION FORMAT: SCAN YOUR HANDWRITTEN ANSWER, MAKE IT A PDF AND NAME IT IN THIS FOLLOWING FORMAT: SECTION_ID_SETA_NAME_Q2.pdf AND UPLOAD IN THE FORM

CSE 221 Quiz 02 B

Time:[7:00pm to 7.40pm] 40 minutes (including submission)

Suppose you are the Mayor of A city Called "Citadel" which has total of 9 central spots[numbering starts from 1 to 9]. Due to some massive break of infectious disease Algid-19 you are sensing the need to lock down the areas to stop the spread of the infection. To achieve that you now have a map of the whole city and how each central area is connected with each other. Keep in mind the connections are made with only 1 way bridge so you cant travel from both side of a bridge. You have to build wall to divide the city into certain areas/hubs where everybody can travel at any places within the hub.



a. Define whether it is a connected or disconnected graph ? [1]

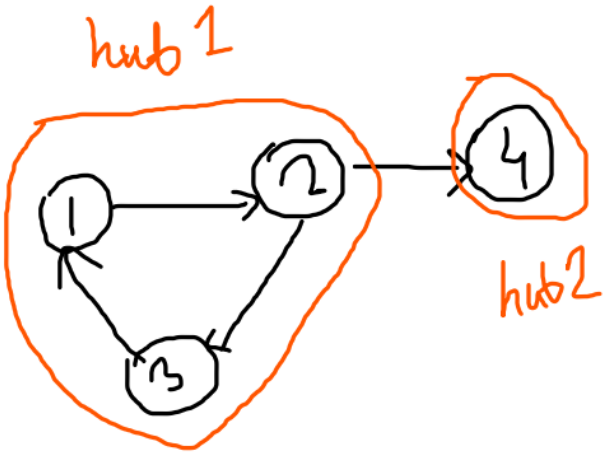
b. Name the algorithm tailored for this situation. [1]

c. Identify the groups of areas/hubs you need to make the boundaries for in-order to contain the infection only within those hubs/areas and It cannot reach other hubs/areas resulting in an outbreak. [a hub can consists of one or more central points]. Give the central spots each hub will have. Show proper simulation of the task using the data structure you are implementing and draw the graph in your script to draw the boundaries which will enclose each hub. [8]

CSE 221 Quiz 02 B

Time:[7:00pm to 7.40pm] 40 minutes (including submission)

For example, 2 hubs can look like this in another city structure: [orange color border is denoted as boundary here just to clarify]



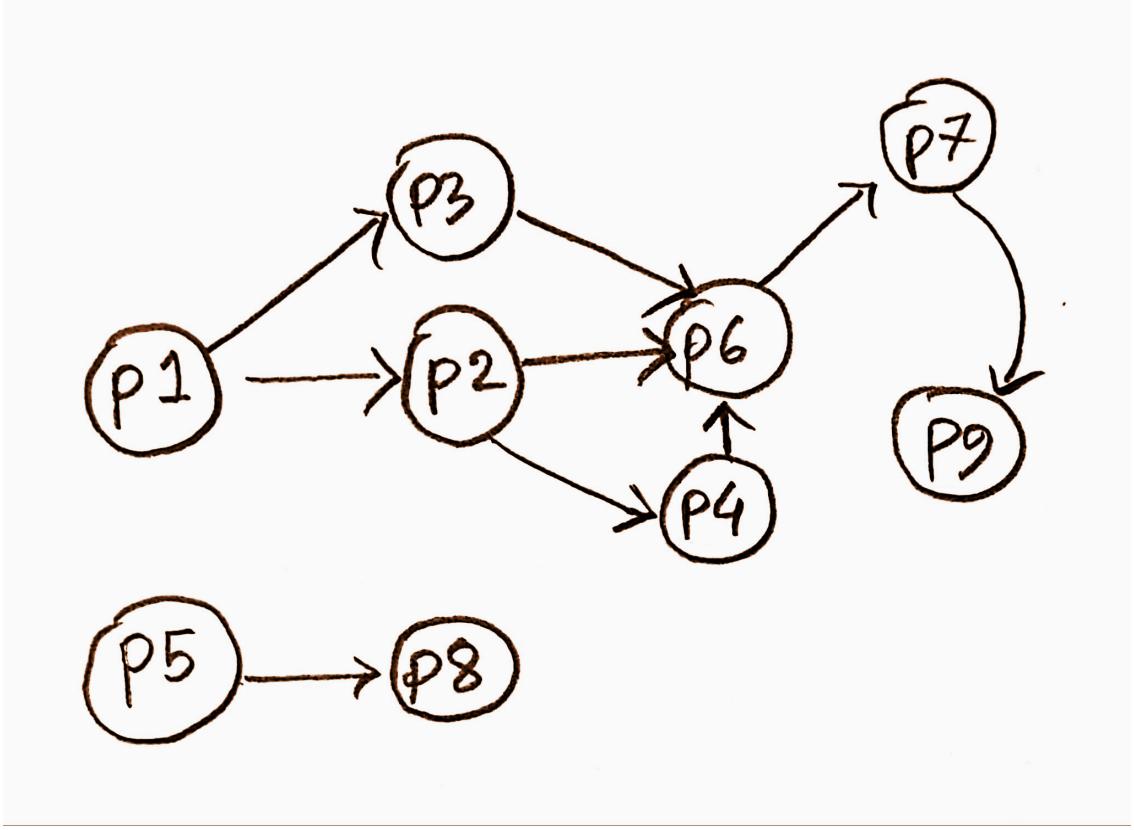
[BONUS] e. If a graph has less amount of adjacent nodes for each node [sparse graph], you are given a source point and a destination point. If the destination is very close from the source. meaning a very small number of hops away from source, which traversal algorithm will work poorly? [1 line answer] [1]

SUBMISSION FORMAT: SCAN YOUR HANDWRITTEN ANSWER, MAKE IT A PDF AND NAME IT IN THIS FOLLOWING FORMAT: SECTION_SETB_ID_NAME_Q2.pdf

CSE 221 Quiz 02 C

Time:40 minutes (including submission)

Suppose you are a system architecture designer working for a company hoping to make a super-computer. With total of 9 processors allocated to you. But the data flow of your scheduling tells you that many processors are dependant on other processors work to be finished in order to begin its own processing. [for example, edge u⇒v means U must be finished first in order for v to start its processing].



a. Name the algorithm tailored for this situation. [1]

b. Manage a sequence of workflows that can be done without breaking any constraints resulting in a successful data transmission which utilizes every processors given to you. You must show proper simulation with step by step decisions you are making and maintaining the data structure you are implementing to achieve your goal [6]

CSE 221 Quiz 02 C

Time:40 minutes (including submission)

c. One day, you see that your intern has accidentally connected the processor 9 directly to the processor 4 (p9→p4) with an additional wire. Briefly explain what will be the consequences of this with logic.

[3]

[BONUS] d. If a graph has less amount of adjacent nodes for each node [sparse graph], you are given a source point and a destination point. If the destination is very close from the source. meaning a very small number of hops away from source, which traversal algorithm will work poorly? [1 line answer] [1]

SUBMISSION FORMAT: SCAN YOUR HANDWRITTEN ANSWER, MAKE IT A PDF AND NAME IT IN THIS FOLLOWING FORMAT: SECTION_SETC_ID_NAME_Q2.pdf